

Exercise 2: Implementing Dependency Injection

Scenario: In the library management application, you need to manage the dependencies between the BookService and BookRepository classes using Spring's IoC and DI.

Code:

applicationContent.xml

```
<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
       xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
       xsi:schemaLocation="
         http://www.springframework.org/schema/beans
         https://www.springframework.org/schema/beans/spring-beans.xsd">

    <bean id="bookRepository"
          class="com.library.repository.BookRepository"/>

    <bean id="bookService"
          class="com.library.service.BookService">

        <property name="bookRepository"
                  ref="bookRepository"/>

    </bean>

</beans>
```

BookRepository.java

```
package com.library.repository;

public class BookRepository {

    public void displayBook() {
        System.out.println("Fetching book details from repository.");
    }

}
```

BookService.java

```
package com.library.service;

import com.library.repository.BookRepository;

public class BookService {

    private BookRepository bookRepository;
```

```

// Setter Injection
public void setBookRepository(BookRepository bookRepository) {
    this.bookRepository = bookRepository;
}

public void displayService() {
    System.out.println("Book Service is working.");
    bookRepository.displayBook();
}
}

```

Output:

The screenshot shows an IDE window with the following components:

- Project Explorer:** Displays the project structure for 'LibraryManagement'. The path 'src > main > java > com > library > service > BookService' is highlighted.
- Editor:** Shows the 'BookService.java' file with the following code:


```

4
5 public class BookService {
6
7     private BookRepository bookRepository;
8
9     // Setter Injection
10    public void setBookRepository(BookRepository bookRepository) {
11        this.bookRepository = bookRepository;
12    }
13
14    public void displayService() {
15        System.out.println("Book Service is working.");
16        bookRepository.displayBook();
17    }
18 }

```
- Run Console:** Shows the output of the application:


```

C:\Program Files\Eclipse Adoptium\jdk-17.0.18.8-hotspot\bin\java.exe" ...
Book Service is working.
Fetching book details from repository.
Process finished with exit code 0

```